

Diffusion in solids

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* What is diffusion?

The mechanism by which matter is transported through matter. It is related to internal atomic movement.

* What conditions must be met for diffusion?

The conditions for diffusion are as follow:

- i) For an atom to make such move there must be an empty adjacent site.
- ii) Atom that is moving should have enough energy to break the bonds with the neighbouring atoms and cause some distortion. The energy is vibrational in nature.

Diffusion Mechanism

There are two main mechanisms of diffusion of atoms in a crystalline lattice.

1. The vacancy or substitutional mechanism
2. The interstitial mechanism

1. **Substitutional diffusion mechanism:** Atoms can move in crystal lattices from one atomic to another if there is enough activation energy provided by the thermal vibration of the atoms and

if there are vacancies or other crystal defects in the lattice for atoms to move into. As the temperature of the metal increases, more vacancies are present and more thermal energy is available and so the diffusion rate is higher at higher temperature.

Interstitial diffusion mechanism: The interstitial diffusion of atoms in crystal lattices takes place when atoms move from one interstitial site to another neighbouring interstitial site without permanently displacing any of the atoms in the matrix crystal lattice. For the interstitial mechanism to be operative the size of the diffusing atoms must be relatively small compared to the matrix atoms. Small atoms such as hydrogen, oxygen, nitrogen and carbon can diffuse interstitially in some metallic crystal lattice.

Diffusion types

There are two types of diffusion.

1. Self diffusion

2. Inter diffusion

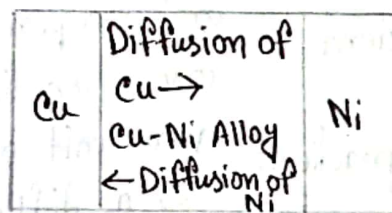
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Self diffusion: The movement of atoms through their own structure is called self diffusion. It is observed in pure metals like Cu, Ni



Inter diffusion: The movement of atoms into different metals from both sides into each other is called inter diffusion. It is observed in binary metal alloy such as Cu-Ni, Cu-Zn etc.



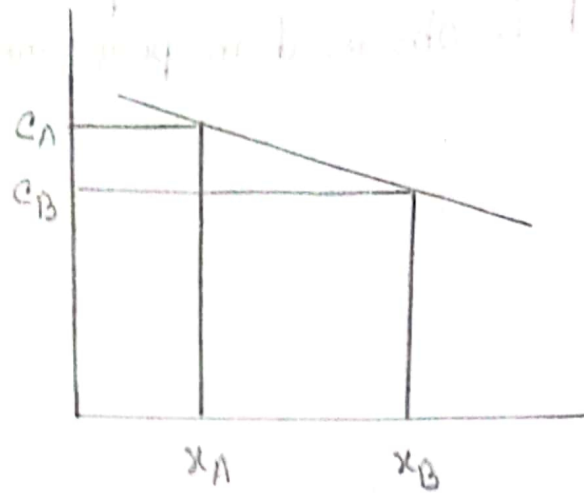
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Steady-state condition

For a diffusing system there is no change in the concentration of the diffusing species with time at different places in the system.

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Diffusion flux



$$J = -D \frac{dc}{dx}$$

Where

J = diffusion flux

D = diffusion coefficient

$\frac{dc}{dx}$ = concentration gradient

The rate of mass transfer and perpendicular to a unit cross sectional area of solid per unit of time expressed as a diffusion flux.

* State the Fick's 1st law of diffusion.

At constant temperature, the flux of diffusing species is directly proportional to the concentration gradient.

$$J \propto \frac{dc}{dx}$$

$$\Rightarrow J = -D \frac{dc}{dx}$$

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Here, the negative sign is used because the diffusion is from a higher to a lower concentration.

This equation states that for steady state equation conditions where

J = Diffusion flux

$$J = \frac{M}{At} = \text{kg/m}^2\text{s or atom/m}^2\text{s}$$

D = Diffusion coefficient/diffusivity

$\frac{dc}{dx}$ = concentration gradient

$$\frac{dc}{dx} = \frac{\Delta c}{\Delta x} = \frac{c_A - c_B}{x_A - x_B}$$

$\therefore$ The concentration gradient is the driving force where driving force is one kind of force that compels a reaction to occur.

$\therefore$ The SI units for this equation are:

$$J \left(\frac{\text{atoms}}{\text{m}^2 \times \text{s}} \right) = D \left(\frac{\text{m}^2}{\text{s}} \right) \frac{dc}{dx} \left(\frac{\text{atoms}}{\text{m}^3} \times \frac{1}{\text{m}} \right)$$

The diffusivity values depend on many variables.



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Diffusion flux (J)

The rate of mass transfer and perpendicular to a unit cross-sectional area of solid per unit of time is expressed as a diffusion-flux. ($\text{kg/m}^2\text{s}$ or $\text{atoms/m}^2\text{s}$)

Diffusion coefficient/diffusivity (D)

A measure of the rate of diffusion in solids at a constant temperature. (m^2/s)

concentration gradient ($\frac{dc}{dx}$)

concentration gradient is the slope at a particular point when concentration c is plotted at a distance x in solid. (kg/m^3 or g/cm^3)

non-steady condition

A diffusing system where the concentration of the diffusing species changes with time at different places in the system.

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* State the Fick's 2nd law of diffusion.

At constant temperature, the rate of change of concentration with time is equal to diffusivity times the rate of change of concentration gradient.

$$\frac{dc}{dt} = \frac{d}{dx} \left(D \frac{dc}{dx} \right)$$

$$\Rightarrow \frac{dc}{dt} = D \frac{d^2c}{dx^2} \quad \text{--- (i)}$$

This equation states that for non-steady state equation conditions.

The boundary condition of Fick's law for non-steady state equation are given below:

1. Before diffusion, any of the diffusing solute atoms in the solid are uniformly distributed with concentration of C_0
2. The value of x at the surface is zero and increases with distance into the solid.
3. The time is taken to be zero at the instant before the diffusion process begins.

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* Why interstitial diffusion is more rapid than vacancy diffusion?

In most metal alloys, interstitial diffusion occurs much more rapidly than diffusion by the vacancy mode because the interstitial atoms are smaller and thus more mobile. Furthermore, there are ~~some~~ more empty interstitial positions than vacancies, hence, the probability of interstitial atomic movement is greater than for vacancy diffusion.

* Discuss the factors that influence diffusion.

The factors that influence diffusion are given below:

1. Type of diffusion mechanism: Whether the diffusion is interstitial or substitutional will affect the diffusivity. Small atoms can diffuse interstitially in the crystal lattice of larger solvent atoms.
2. Temperature: Temperature at which the diffusion takes place greatly affects the value of the diffusivity. As the temperature is increased, the diffusivity also increases.

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2. Temperature: Temperature at which the diffusion takes place greatly affects the value of the diffusivity. As the temperature is increased, the diffusivity also increases.

3. Type of crystal structure: Type of crystal structure of the solvent lattice is important. For example, the diffusivity of carbon in BCC iron is $10^{-12} \text{ m}^2/\text{s}$ at 500°C , which is much greater than $5 \times 10^{-15} \text{ m}^2/\text{s}$, the value for the diffusivity of carbon in FCC iron at the same temperature. The reason for this difference is that the BCC crystal structure has a lower atomic packing factor compared to that of the FCC crystal structure.

4. Type of crystal imperfections: Type of crystal imperfections present in the region of solid-state diffusion is also important. More open structures allow for more rapid diffusion of atoms. For example, diffusion takes place more rapidly along grain boundaries than in the grain matrix in metals and ceramics. Excess vacancies will increase diffusion rates in metals and alloys.

5. Concentration of the different species: In that higher concentrations of diffusing solute atoms will affect the diffusivity. This aspect of solid-state diffusion is very complex.

Crystalline Imperfections

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In reality, crystals are never perfect and contain various types of imperfections and defects that affect many of their physical and mechanical properties, which in turn affect many important engineering properties of materials.

Crystal lattice imperfections are classified according to their geometry and shape. The three main divisions are

1. zero-dimensional or point defects
2. one-dimensional or line defects
3. Two-dimensional defects or planar defects

1. point defects: The simplest point defect is the vacancy, an atom site from which an atom is missing. Vacancies may be produced during solidification as a result of local disturbances during the growth of crystals, or they may be created by atomic rearrangements in an existing crystal due to atomic mobility.

2. Line defects: Line imperfections or dislocations, in crystalline solids are defects that cause lattice distortion centered around a line. Dislocations are created during the solidification of crystalline solids.

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3. Planar defects : Planar defects include external surfaces, grain boundaries, twins, low-angle boundaries, high-angle boundaries, twists and stacking faults. The free or external surface of any material is the most common type of planar defect. External surfaces are considered defects because the atoms on the surface are bonded to other atoms only on one side.

*** Volume defects :** Volume or three-dimensional defects form when a cluster of point defects join to form a three-dimensional void or a pore. Conversely, a cluster of impurity atoms may join to form a three-dimensional precipitate. The size of a volume defect may range from a few nanometers to centimeters or something larger. Such defects have a tremendous effect or influence on the behavior and performance of the material. Finally, the concept of a three-dimensional or volume defect may be extended to an amorphous region within a polycrystalline material. Such materials were briefly

Effect of temperature on diffusion in solids

Since atomic diffusion involves atomic movements, it is to be expected that increasing the temperature of a diffusion system will increase the diffusion rate. By experiment, it has been found that the temperature dependence of the diffusion rate of many diffusion systems can be expressed by the following Arrhenius-type equation:

$$D = D_0 e^{-\frac{Q}{RT}}$$

where,

D = diffusivity, m^2/s

D_0 = ~~pro~~ proportionally constant, m^2/s , independent of temperature in range for which equation is valid

Q = activation energy of diffusing species, J/mol or cal/mol

R = molar gas constant = $8.314 \text{ J/(mol}\cdot\text{K)}$ or $1.987 \text{ cal/(mol}\cdot\text{K)}$

T = Temperature, K

$$\ln D = \ln D_0 - \frac{Q}{RT}$$

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Some definitions

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Activation energy: The additional energy required above the average energy for a thermally activated reaction to take place.

Volume diffusion: Atomic migration in the grain interiors of a polycrystalline aggregate.

Grain boundary diffusion: Atomic migration at the grain boundaries of a polycrystalline aggregate.

* What is the driving force for steady state diffusion?

Ans: The driving force for steady state diffusion is the existence of a concentration gradient, which is the difference in concentration between two regions.

* Other diffusion paths

Atomic migration may also occur along dislocations, grain boundaries, and external surfaces. These are sometimes called short-circuit diffusion paths in as much as rates are much faster than for bulk diffusion. However, in most situations

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short circuit contributions to the overall diffusion flux are insignificant because the cross-sectional areas of these paths are extremely small.

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